Peinan **Li**

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Robotics, embodied intelligence, autonomous driving, multimodal AI, and full-stack engineering across mechanics, electronics, algorithms, and AI.

Summary

Undergraduate student in Mechatronics Engineering at Tongji University, focused on robotics, embodied intelligence, autonomous driving, multimodal reasoning, and interpretable AI systems. Experienced across mechanics, electronics, algorithms, and AI engineering, with hands-on work in robotics competitions, research prototypes, autonomous driving algorithms, and early-stage product development.

Education

School of Mechanical Engineering and Robotics, Tongji University | 2023.09 - 2027.07 | **B.Eng. in Mechatronics Engineering**

Coursework and project work across robotics, intelligent systems, mechatronics, and AI engineering.
Active in innovation competitions, robotics team rebuilding, research assistance, and engineering prototypes.

Research Experience

AutoLab, School of Artificial Intelligence, Shanghai Jiao Tong University | 2026.01 - Present | **Research Assistant**

Worked on autonomous driving, embodied intelligence, multimodal reasoning, and interpretable AI systems.
Represented AutoLab at the CGIS experts visit to Yinwang, joining exchanges on intelligent mobility and frontier engineering practice.

Entrepreneurship

Xingzhou Yinyue (Shanghai) Embodied Intelligent Robotics Co., Ltd. | Present | **CTO**

Led technical planning and engineering implementation for embodied intelligent robotics products.
Drove integration of robotics, AI algorithms, and software-hardware systems toward real product scenarios.

Internship Experience

Chongqing Shiwei Data Technology Co., Ltd. · NUS Collaboration | 2025.08 - 2026.01 | **Algorithm Engineer**

Led DP180 multi-camera loop-closure localization algorithm development for robust localization in complex scenes.
Led long-range LiDAR 3D mapping algorithm development for multi-sensor spatial perception and map construction.

Hello Group, Shanghai Zaofu Technology Co., Ltd. | 2026.02 - 2026.05 | **End-to-End Algorithm Engineer Intern**

Worked on end-to-end algorithm engineering, focusing on models, data, and practical engineering pipelines.

Research Publications

Neuro-Causal Drive: Interpretable Traffic Accident Analysis | ICME 2026 · Accepted | **First Author, Oral**

Independently wrote the paper and proposed a neuro-symbolic framework for long-horizon, interpretable accident analysis.

Evidential-LIO: Uncertainty-Aware Dynamic Mapping | IROS 2026 · Submitted | **Submitted**

DriveNarrate: Decision Bottlenecks for Driving | ACM MM 2026 · Submitted | **Submitted**

Multimodal Learning for Defect Explanation | ACM MM 2026 · Submitted | **Submitted**

Context Paging for Long-Horizon LLM Agents | COLM 2026 · Submitted | **Submitted**

FlexPlanner: Adaptive Planner Selection for Driving | NeurIPS 2026 · Submitted | **Submitted**

Project Experience

Built interdisciplinary projects across mechanics, electronics, and AI, covering robotic systems, autonomous driving algorithms, localization and mapping, multimodal AI, and intelligent equipment prototypes. Experienced in turning complex engineering ideas into real systems through concept design, software-hardware integration, competition validation, and product-oriented exploration.

Competitions & Honors

2025 Science and Innovation Star, School of Mechanical Engineering and Robotics, Tongji University; only 10 selected	2025	**Science and Innovation Star**
2025 China University Student Mechanical Engineering Innovation and Creativity Competition	2025	**Second Prize**
14th Shanghai University Student Engineering Practice and Innovation Ability Competition	2025	**Second Prize**
12th Shanghai University Student Mechanical Engineering Innovation Competition	2024	**Second Prize**
2025 China University Student Mechanical Engineering Innovation and Creativity Competition, Logistics Technology Track	2025	**Third Prize**
6th Global Campus Artificial Intelligence Algorithm Elite Competition	2024	**Third Prize**
2nd Shanghai University Student Energy Conservation and Emission Reduction Competition	2025	**Second Prize**
2024 Intelligent Robot Innovation and Entrepreneurship Competition	2024	**First Prize**
RAICOM Shanghai Division ROS Robot Virtual Simulation Competition	2024	**Third Prize**
Tongji University Challenge Cup Selection	2025	**First Prize**
China International College Students Innovation Competition, Tongji Campus Final	2025	**Gold Award**

Patents

Force-controlled mechanically lockable electric push rod; Application No. 202510963355.9	2025.07.11	**Invention Patent Accepted, First Inventor**
Parallel end-effector multi-degree-of-freedom robotic arm; Application No. 202521461640.2	2025.07.11	**Utility Model Patent Accepted, First Inventor**
Human-assistive robotic arm, mobility assistance control method, and intelligent wheelchair; Application No. 202510931694.9	2025.07.07	**Invention Patent Accepted, First Inventor**

Special Experiences & Skills

Tongji University Robocon Robotics Team 2024.03 - Present **Captain**

Rebuilt the team, organized technical materials, designed competition strategies, and trained core members.
Worked across mechanical design, electrical control, algorithms, and competition execution.

XbotPark Ningbo Base x Ningbo Institute of Intelligent Technology Innovation Camp Winter Camp **Participant**

Presented project ideas and joined exchanges on robotics, intelligent technology, and entrepreneurship practice.

CGIS Experts Visit Yinwang Special Event May 16 **AutoLab Representative**

Represented AutoLab in exchanges on intelligent mobility, autonomous driving, and frontier engineering practice.

Robotics, Algorithms, and Full-Stack Engineering Ongoing **Integrated Skills**

Robotics: mechanical design, electrical control, Robocon strategy, team building, and member training.
Algorithms: localization, mapping, LiDAR 3D reconstruction, multi-camera loop closure, autonomous driving, and multimodal reasoning.
Engineering: full-stack development, AI applications, research prototypes, and early-stage product planning.